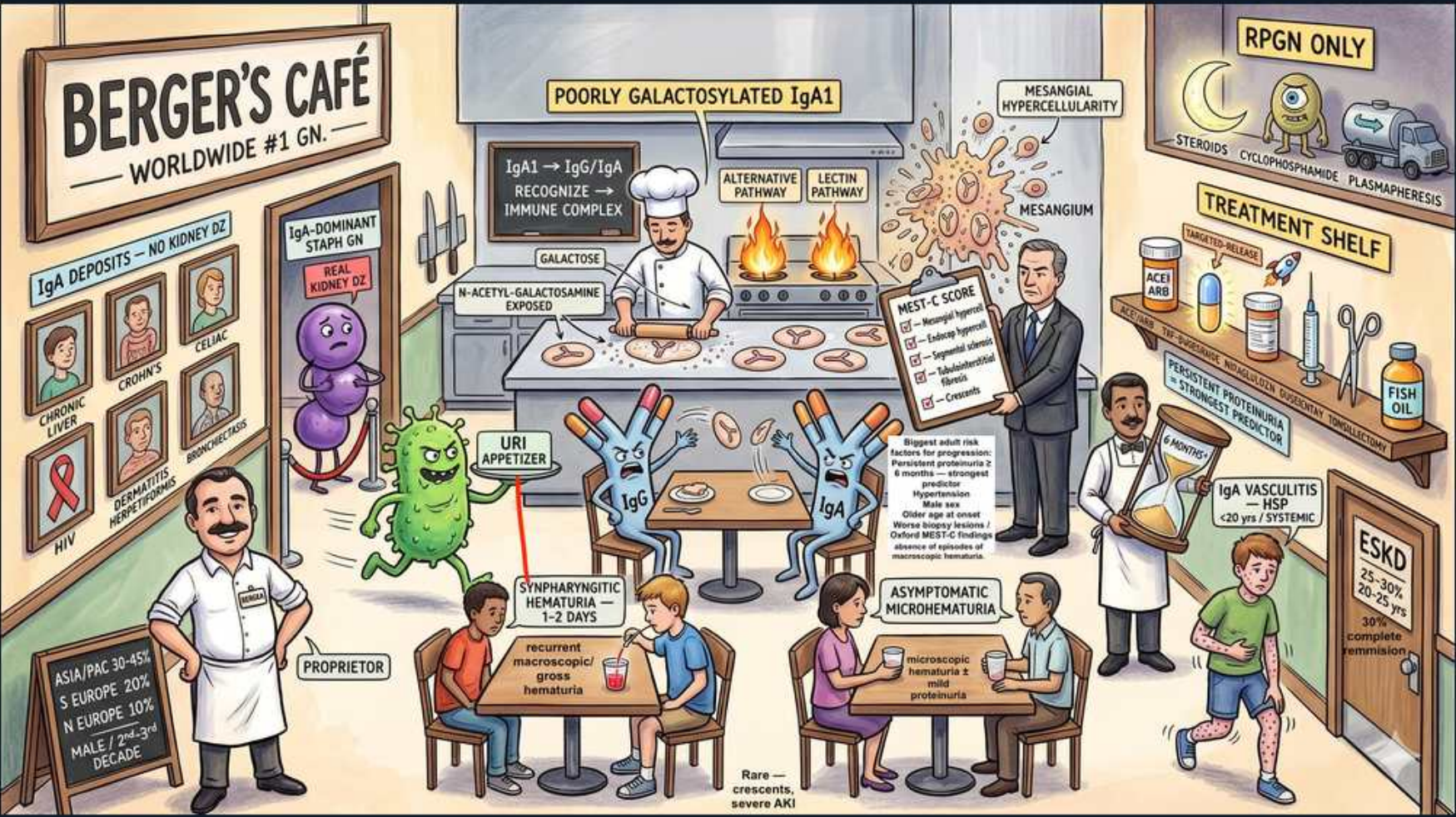


Glomerular (Nephritic) — IgA Nephropathy (Berger's Disease)



1. Berger's Café

WHAT YOU SEE

A storefront sign reading BERGER'S CAFÉ — WORLDWIDE #1 GN — the named café is the mnemonic anchor for Berger's disease, the commonest GN worldwide.

WHAT IT ENCODES

IgA nephropathy (Berger's disease) is the most common primary glomerulonephritis worldwide, with highest prevalence in East Asians and whites and notably lower rates in those of African descent. It typically presents in the second-to-third decade, more often in males. Mnemonic: 'IgA = I Got Asia,' and 'Berger = the burger you eat right when you get a cold' (synpharyngitic).

BOARD TRAP

PSGN is the most common GN worldwide — IgA nephropathy holds that title; PSGN is instead the classic post-infectious GN of children weeks after strep

2. Poorly galactosylated IgA1

WHAT YOU SEE

A banner reading POORLY GALACTOSYLATED IgA1 with a chef exposing N-ACETYL-GALACTOSAMINE — the chef stripping the sugar off depicts galactose-deficient IgA1, the core defect.

WHAT IT ENCODES

The core defect is overproduction of galactose-deficient IgA1, which the immune system recognizes as an autoantigen, generating anti-glycan IgG/IgA antibodies that form circulating immune complexes. These complexes deposit in the mesangium and trigger inflammation. This 'multi-hit' model (abnormal IgA1 → autoantibody → immune complex → mesangial deposition → injury) is the high-yield mechanism.

3. Mesangial IgA deposits

WHAT YOU SEE

Café portraits of IgA-deposit patients beside a crossed-out NO KIDNEY/STAPH sign and chefs at a MESANGIAL station — the IgA being plated into the mesangium IS the dominant mesangial IgA on IF.

WHAT IT ENCODES

The diagnostic hallmark on immunofluorescence is dominant/co-dominant IgA deposition in the MESANGIUM (often with C3 and IgG), and EM shows mesangial electron-dense deposits. Light microscopy reveals mesangial hypercellularity. Diagnosis ultimately requires biopsy because serum IgA levels are unreliable (elevated in only ~50%). Mesangial IgA is the single best discriminator.

BOARD TRAP

Subepithelial 'humps' on EM — those are PSGN; IgA nephropathy deposits are MESANGIAL, not subepithelial

4. Immune complex / alt + lectin pathway

WHAT YOU SEE

A stovetop with IgA1 + IgG/IgA forming an immune complex over ALTERNATIVE and LECTIN PATHWAY flames — the two burning pathways depict complement activated locally (yet serum complement stays normal).

WHAT IT ENCODES

The mesangial immune complexes activate complement via the ALTERNATIVE and LECTIN pathways (not the classical pathway), yet serum complement remains NORMAL because consumption is localized rather than systemic. This is why IgA is a NORMAL-complement nephritic disease, unlike PSGN, lupus, MPGN, and endocarditis. Normal complement + IgA on biopsy is the combination to remember.

BOARD TRAP

Serum C3 is low in IgA nephropathy — it is NORMAL; low complement instead points to PSGN, lupus, MPGN, or endocarditis

5. Mesangial hypercellularity

WHAT YOU SEE

A burst of proliferating cells labelled MESANGIAL HYPERCELLULARITY next to a MEST-C SCORE board — the crowded mesangium IS the hypercellularity the Oxford score grades.

WHAT IT ENCODES

Light microscopy shows mesangial proliferation/hypercellularity. The Oxford MEST-C classification (Mesangial hypercellularity, Endocapillary hypercellularity, Segmental sclerosis, Tubular atrophy/interstitial fibrosis, Crescents) scores biopsy features and predicts prognosis and treatment response. Tubular atrophy/fibrosis (T) and crescents (C) are the strongest predictors of progression.

6. Synpharyngitic hematuria

WHAT YOU SEE

A sick diner with cola-colored urine tagged SYNPHARYNGITIC HEMATURIA 1-2 DAYS — the hematuria appearing right with the infection IS the synpharyngitic timing (vs PSGN's weeks-later lag).

WHAT IT ENCODES

The classic presentation is gross (cola-colored) hematuria appearing WITHIN 1–2 DAYS of a mucosal infection — 'synpharyngitic,' meaning concurrent with the pharyngitis. This tight temporal link is the key timing discriminator from PSGN, which lags the infection by weeks. A young adult with recurrent visible hematuria during URIs is the prototypical vignette.

BOARD TRAP

Hematuria appearing 2–3 weeks AFTER a sore throat — that delayed pattern is PSGN; IgA hematuria is synpharyngitic (1–2 days, concurrent)

7. URI appetizer

WHAT YOU SEE

An URI APPETIZER plate served to the IgA diners — the mucosal infection on the menu is what triggers the IgA flare.

WHAT IT ENCODES

A mucosal infection — upper respiratory or gastrointestinal — provokes the IgA response and triggers the flare, because IgA is the predominant mucosal immunoglobulin. This mucosal trigger explains the synpharyngitic timing and the recurrent, relapsing-with-infections course. Tonsillitis and GI infections are common precipitants.

8. Asymptomatic microhematuria

WHAT YOU SEE

A well-looking diner labelled ASYMPTOMATIC MICROHEMATURIA — depicts the common incidental finding of persistent microscopic hematuria with preserved function.

WHAT IT ENCODES

Beyond the dramatic gross-hematuria episodes, the most common presentation overall is incidentally discovered persistent microscopic hematuria with or without mild (sub-nephrotic) proteinuria and preserved renal function. Many such patients are found on routine urinalysis. Dysmorphic RBCs and RBC casts indicate the glomerular (nephritic) origin.

9. RPGN only (rare crescents)

WHAT YOU SEE

A truck/sign reading RPGN ONLY loaded with STEROIDS / CYCLOPHOSPHAMIDE / PLASMAPHERESIS — the heavy artillery is reserved ONLY for the rare crescentic RPGN subset.

WHAT IT ENCODES

A crescentic, rapidly progressive course is RARE in IgA nephropathy; only this severe subset warrants aggressive immunosuppression (high-dose corticosteroids ± cyclophosphamide, occasionally plasmapheresis). Routine IgA does NOT get cytotoxic therapy. Reserve immunosuppression for crescentic RPGN on biopsy or rapidly declining GFR despite supportive care.

BOARD TRAP

All IgA patients need cyclophosphamide — most do not; cytotoxic immunosuppression is reserved for the rare crescentic/RPGN subset

10. Treatment shelf — ACEi/ARB + fish oil

WHAT YOU SEE

A TREATMENT SHELF stocked with ACEi/ARB and a FISH OIL bottle — the supportive first-line therapy (RAAS blockade + fish oil) for typical IgA.

WHAT IT ENCODES

First-line therapy for the typical patient is supportive: an ACE inhibitor or ARB to control blood pressure and reduce proteinuria (target proteinuria below ~1 g/day), with fish oil as an adjunct. Corticosteroids are considered when proteinuria persists above ~1 g/day despite optimized RAAS blockade. SGLT2 inhibitors are increasingly used to slow CKD progression. Escalate to immunosuppression only for high-risk/progressive disease.

11. IgA vasculitis (HSP) link

WHAT YOU SEE

A child with purpuric lower-leg spots tagged IgA VASCULITIS — HSP, SYSTEMIC — the palpable purpura depicts HSP as the systemic form of the same IgA disease.

WHAT IT ENCODES

Henoch-Schönlein purpura, now called IgA vasculitis, is the SYSTEMIC form of the same disease — identical IgA-dominant immune deposits but with extrarenal involvement: palpable purpura on the lower extremities/buttocks, arthralgias, abdominal pain (GI bleeding/intussusception), and nephritis. It is the most common childhood vasculitis and usually self-limited. Same renal histology as Berger's, broader clinical syndrome.

12. ESKD long-term

WHAT YOU SEE

A dialysis/ESKD sign reading 20-40% — flags the long-term progression risk to end-stage kidney disease.

WHAT IT ENCODES

IgA nephropathy is NOT uniformly benign: roughly 20–40% of patients progress to end-stage kidney disease over 10–20 years. Poor prognostic markers include hypertension, persistent proteinuria above ~1 g/day, reduced GFR at diagnosis, and adverse MEST-C features (T and C). It can also recur in a transplanted kidney, though graft loss from recurrence is uncommon.